Topic 4 — Digestion, Respiration and Excretion

Covers syllabus (click to expand)

Digestive System Anatomy (Mouth, Teeth, Salivary Glands, Stomach, Liver, Gall Bladder, Pancreas, Small & Large Intestine) | Dental Formula & Tooth Structure (Enamel) | Digestive Enzymes & Juices (Ptyalin, Pepsin, Rennin, Bile, Trypsin, Lipase, Succus Entericus) | Digestion of Carbohydrates, Proteins, Fats, Starch & Cellulose | Absorption Mechanics (Villi, Lacteals, Chylomicrons) | Respiratory System (Airway Anatomy, Larynx, Trachea, Alveoli) | Breathing Mechanics (Diaphragm, Intercostal Muscles) & Lung Volumes (Tidal, Vital Capacity, Residual) | Gas Transport (O_2 Oxyhaemoglobin, CO_2 Bicarbonate, Bohr Effect) & Carbon Monoxide Poisoning | Cellular Respiration (Glycolysis, Krebs Cycle, ETS, ATP Yield, Lactic Acid Debt) | Excretory System Anatomy (Kidney, Nephron Structure, Malpighian Body, Tubules) | Urine Formation (Ultrafiltration, Selective Reabsorption, Tubular Secretion) | Counter-Current Mechanism & Hormonal Regulation (ADH, Aldosterone, RAAS, ANF) | Nitrogenous Wastes (Ammonotelic, Ureotelic, Uricotelic) | Excretory Disorders (Renal Calculi, Uremia, Gout, Dialysis)

Sources baked in: NCERT Class 10 & 11 (Life Processes, Digestion and Absorption, Breathing and Exchange of Gases, Excretory Products and Elimination), Lucent General Science (Physiology & Enzymes), UPPCS Prelims PYQs (2018–2025) and UKPCS Prelims **Weight:** ★★★ High — heavily tested: site of initiation of protein digestion, stomach vs pancreatic enzymes, bile function, lung volumes, gas transport forms, nephron reabsorption sites, and nitrogenous waste types **Last verified:** September 2026

Current Affairs

Year	Event / Fact	Must-Score Fact
2024	UKPCS tested agent class utilized as an **immunosuppressor in kidney transplantation**.	Keyed answer: **Monoclonal antibodies** (e.g. Basiliximab / Daclizumab) and calcineurin inhibitors (Cyclosporine A), preventing allograft organ rejection; antibiotics do not prevent graft rejection.
Still Current	**Kidney stone composition in India**: Over **80% of urinary calculi** are composed of insoluble **Calcium Oxalate (CaC_2O_4) crystals**, followed by calcium phosphate and uric acid.	Dietary facts: Foods rich in oxalates (tomatoes, spinach, beetroot, chocolate) accelerate stone formation in genetically prone individuals.
Still Current	Medical device fact: **Haemodialysis machine ("Artificial Kidney")** operates on the physical principle of **Dialysis across a semi-permeable cellophane membrane**.	Patient's blood is drawn from an artery, mixed with anticoagulant **Heparin**, filtered against dialyzing fluid, rewarmed, and reinfused with **Anti-heparin** into a vein.

4.1 Digestive Physiology: From Mouth to Absorption

- **The Human Mouth & Mechanical Mastication**:
- Food is ingested into the buccal cavity, where it is mechanically masticated by teeth and lubricated by saliva to form a swallowed mass called a **Bolus**.
- **Human Dentition Characteristics**:
  - **Thecodont**: Every tooth is embedded in a deep bony jaw socket (alveolus).
  - **Diphyodont**: Two sets of teeth during lifetime: deciduous / milk teeth (20 teeth) replaced by permanent adult teeth (32 teeth).
  - **Heterodont**: Four morphologically and functionally distinct types of teeth:
  - **Incisors (I)**: Chisel-shaped, for biting and cutting food.
  - **Canines (C)**: Dagger-shaped, sharp, for tearing meat (prominent in carnivores; absent in rodents → toothless gap termed **Diastema**).
  - **Premolars (PM)**: Bicuspid, for grinding and chewing food (absent in milk dentition).
  - **Molars (M)**: Broad, multicuspid, for crushing and grinding food; the 3rd molar on each side is the **Wisdom Tooth** (erupts at age 17–25; vestigial).
- **Salivary Glands**: Three pairs of exocrine salivary glands secrete 1.0–1.5 litres of saliva per day:
  1. **Parotid Glands**: Largest pair, situated below and in front of the ears; opens via **Stensen's duct**; infection by Paramyxovirus causes painful swelling called **Mumps**.
  2. **Submandibular / Submaxillary Glands**: Located at the angles of the lower jaw; opens via **Wharton's duct**; secretes the majority of resting saliva.
  3. **Sublingual Glands**: Smallest pair, situated under the tongue; opens via **Ducts of Rivinus**. - Saliva contains **Ptyalin (Salivary Amylase)** which digests cooked starch into maltose, **Lysozyme** (bactericidal enzyme cleaving bacterial cell walls), secretory **IgA**, and bicarbonate buffer (pH 6.8).
- **Gastric Digestion in the Stomach**:
- The stomach is a J-shaped muscular organ with 4 regions: Cardiac, Fundus, Body, and Pylorus. Guarded by the **Gastro-oesophageal sphincter** at the entry and the **Pyloric sphincter** at the exit.
- Gastric mucosa contains branched tubular **Gastric Glands** possessing three major cell types:
  1. **Mucous Neck Cells**: Secrete protective alkaline mucus.
  2. **Peptic / Chief / Zymogenic Cells**: Secrete inactive proenzymes **Pepsinogen** and **Prorennin**.
  3. **Parietal / Oxyntic Cells**: Secrete **Hydrochloric Acid (HCl)** and **Castle's Intrinsic Factor** (indispensable for Vitamin B12 absorption in the ileum).
- Functions of Gastric HCl: Creates a strongly acidic environment (pH 1.5–2.0); kills ingested food-borne bacteria; decalcifies bone fragments; converts inactive pepsinogen into active **Pepsin**.

- **Pepsin** cleaves internal peptide bonds of dietary proteins, converting complex proteins into soluble **Proteoses and Peptones**.
- Food is churned into an acidic, semi-fluid pulpy mass called **Chyme**.
- **Duodenal Digestion: Liver, Bile & Pancreas**:
- Acid chyme enters the C-shaped **Duodenum** through the pyloric valve; triggers secretion of two local enterogastrone hormones:
  - **Secretin**: Discovered by **Bayliss and Starling (1902)** (the **first discovered hormone in history**); stimulates pancreatic secretion of watery bicarbonate juice to neutralise gastric acid.
  - **Cholecystokinin (CCK / Pancreozymin)**: Stimulates gall bladder contraction to eject bile, and stimulates pancreatic secretion of digestive enzymes.
- **The Dual Pancreas**:
  - Endocrine Islets of Langerhans (1–2% mass): Insulin and Glucagon (Topic 3).
  - Exocrine Acinar Tissue (98% mass): Secretes complete pancreatic juice into the duodenum via the **Pancreatic Duct (Duct of Wirsung)**, joining the common bile duct to form the **Hepato-pancreatic Ampulla (Ampulla of Vater)**, guarded by the **Sphincter of Oddi**.
- **Intestinal Absorption & The Hepatic Portal Axis**:
- The small intestine (Jejunum and Ileum) completes digestion via brush-border enzymes of the **Succus Entericus**.
- Monosaccharides (glucose, galactose) and amino acids are actively transported across villus enterocytes into local blood capillaries; coalesce into the **Hepatic Portal Vein**, which transports newly absorbed nutrients directly to the **Liver** for metabolic sorting, storage, and detoxification before systemic circulation.
- Long-chain fatty acids and fat-soluble vitamins cannot enter blood capillaries directly; packaged into protein-jacketed **Chylomicrons**, absorbed into central lymphatic **Lacteals**, and drained via the **Thoracic Lymphatic Duct** into the left subclavian vein.

Bilingual Terminology -- Digestion, Respiration and Excretion

English Term	Hindi Term	Key Anchor
Digestion (पाचन)	पाचन	Breaking down food into absorbable molecules
Alimentary Canal (आहार नाल)	आहार नाल / पाचन नाल	Mouth to anus; complete digestive tract
Saliva (लार)	लार	Secretion of salivary glands; contains **salivary amylase** (ptyalin)
Stomach (आमाशय)	आमाशय	Secretes gastric juice (HCl + pepsin + mucin)
Small Intestine (छोटी आंत)	क्षुद्रांत्र	Main site of digestion and absorption; duodenum, jejunum, ileum
Large Intestine (बड़ी आंत)	वृहदांत्र	Water absorption; formation of faeces; colon, rectum
Liver (यकृत)	यकृत	Largest gland; secretes bile; detoxification; glycogen storage
Pancreas (अग्न्याशय)	अग्न्याशय	Secretes pancreatic juice (amylase, lipase, trypsinogen); also insulin/glucagon
Bile (पित्त)	पित्त	Emulsifies fats; secreted by liver; stored in gallbladder
Villi (आंत्र अंकुर)	रसांकुर	Finger-like projections in small intestine; increase surface area for absorption
Respiration (श्वसन)	श्वसन	Process releasing energy from food; cellular respiration = aerobic/anaerobic
Aerobic Respiration (वायवीय श्वसन)	वायवीय श्वसन	Uses O2; produces CO2 + H2O + 38 ATP per glucose
Anaerobic Respiration (अवायवीय श्वसन)	अवायवीय श्वसन	No O2; produces lactic acid (muscles) or ethanol+CO2 (yeast)
Alveoli (वायुकोश)	वायुकोश	Tiny air sacs in lungs; site of gas exchange (O2 in, CO2 out)
Diaphragm (मध्यपट)	मध्यपट	Dome-shaped muscle below lungs; main breathing muscle
Excretion (उत्सर्जन)	उत्सर्जन	Removal of metabolic waste products from the body
Kidney (वृक्क)	वृक्क / गुर्दा	Primary excretory organ; filters blood to form urine
Nephron (नेफ्रोन)	वृक्काणु / नेफ्रोन	Structural and functional unit of kidney; ~1 million per kidney
Urea (यूरिया)	यूरिया	Main nitrogenous waste in mammals; formed in liver
Uric Acid (यूरिक अम्ल)	यूरिक अम्ल	Nitrogenous waste in birds, reptiles, insects (less water loss)
Dialysis (डायलिसिस)	डायलिसिस	Artificial kidney function; filters blood when kidneys fail

4.2 Respiratory Mechanics & Gas Transport Dynamics

- **The Respiratory Pathway Anatomy**:
- External nostrils → Nasal cavity (lined by vascular pseudostratified ciliated columnar epithelium and olfactory Schneiderian membrane) → Pharynx (common passage for food and air) → Larynx (guarded by the elastic cartilaginous Epiglottis) → Trachea (C-shaped cartilage rings) → Primary Bronchi → Secondary/Tertiary Bronchi → Terminal Bronchioles → Respiratory Bronchioles → Alveolar Ducts → **Alveoli**.
- **The Alveolar Respiratory Membrane**:
  - Ultra-thin diffusion barrier (< 0.5 microns thick) composed of: (a) Alveolar type-I squamous epithelium, (b) Epithelial basement membrane, (c) Capillary basement membrane, and (d) Capillary endothelial wall.

- Alveolar type-II cells secrete **Pulmonary Surfactant (Dipalmitoyl lecithin)**; lowers surface tension of alveolar fluid, preventing lung collapse during expiration; premature neonates lacking surfactant develop fatal **Infant Respiratory Distress Syndrome (IRDS)**.

- **Physical Chemistry of Gas Transport**:
- **Oxygen (O_2) Exchange**:
  - Alveolar partial pressure of oxygen (pO_2) is high (~$104\ \text{mm Hg}$), while deoxygenated venous capillary pO_2 is low (~$40\ \text{mm Hg}$); O_2 diffuses rapidly into blood.
  - Haemoglobin consists of globin protein and 4 haem iron groups; binds oxygen cooperatively, producing a characteristic **Sigmoidal Oxyhaemoglobin Dissociation Curve**.
  - **Right Shift of Dissociation Curve (Facilitates O_2 Release at Tissues)**: Caused by elevated pCO_2, elevated temperature, elevated 2,3-diphosphoglycerate (2,3-DPG), and decreased blood pH / acidosis (**The Bohr Effect**).
  - **Left Shift of Dissociation Curve (Increases O_2 Affinity)**: Caused by low pCO_2, high pH (alkalosis), and lower temperature (as seen in alveolar capillaries).

- **Carbon Dioxide (CO_2) Exchange**:
  - Tissue pCO_2 is high (~$45\ \text{mm Hg}$); diffuses into blood capillaries where pO_2 is lower (~$40\ \text{mm Hg}$).
  - **Bicarbonate Mechanism (70%)**:

$$CO_2 + H_2O \xrightleftharpoons{\text{Anhydrase}} \text{Carbonic} H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

  - As HCO_3^- ions accumulate inside erythrocytes, they diffuse down their concentration gradient into blood plasma; to preserve electrical neutrality, chloride ions (Cl^-) diffuse from plasma into the erythrocyte — this phenomenon is the **Chloride Shift / Hamburger Phenomenon**.

- **Cellular Respiration Pathways & Energetics**:
- **Phase 1: Glycolysis (EMP Pathway)**:
  - Occurs in the **cytoplasm**; anaerobic (does not require O_2); common to both aerobic and anaerobic organisms.
  - Converts 1 molecule of Glucose ($6C$) into 2 molecules of Pyruvate ($3C$); yields net **2 ATP** (substrate-level phosphorylation) and **2 NADH**.

- **Phase 2: Link Reaction (Oxidative Decarboxylation)**:
  - Pyruvate enters the **mitochondrial matrix**; converted to Acetyl-CoA by Pyruvate Dehydrogenase, releasing CO_2 and yielding 2 NADH.

- **Phase 3: Krebs Cycle (Citric Acid / TCA Cycle)**:
  - Occurs in the **mitochondrial matrix**; discovered by Sir Hans Krebs (Nobel 1953).
  - Acetyl-CoA ($2C$) combines with Oxaloacetate ($4C$) to form Citrate ($6C$); cyclic oxidation releases CO_2, yielding **2 ATP / GTP**, **6 NADH**, and **2 FADH2**.

- **Phase 4: Oxidative Phosphorylation (ETS)**:
  - Inner mitochondrial membrane cristae; electrons from NADH and FADH2 pass through cytochromes to molecular oxygen (terminal electron acceptor, forming H_2O); drives proton gradient pumping through $F_0 - F_1$ complexes to generate ATP.
  - Total net yield: **36 to 38 ATP per molecule of glucose**.

4.3 Renal Excretion, Nephron Mechanics & Water Balance

- **Gross Anatomy of the Human Urinary System**:
- Pair of bean-shaped reddish-brown kidneys located retroperitoneally between T12 and L3 vertebrae; right kidney is positioned slightly lower than left due to the massive liver above it.
- External notch called the **Hilum** admits renal artery, renal vein, nerves, and the **Ureter**.
- Internally divided into an outer pale **Cortex** and inner dark **Medulla** consisting of 8–18 conical **Renal Pyramids** projecting into minor and major calyces, opening into the funnel-shaped **Renal Pelvis**.
- **Ultrastructure of the Nephron**:
- Two primary anatomical divisions:

1. **Renal Corpuscle (Malpighian Body)**: Located in the cortex. Composed of the **Glomerulus** (a high-pressure tuft of non-anastomosing capillaries supplied by a wide **Afferent Arteriole** and drained by a narrower **Efferent Arteriole**) enclosed within **Bowman's Capsule**. - Inner visceral wall of Bowman's capsule is lined by specialized epithelial cells called **Podocytes**, whose pedicels interdigitate to leave narrow **Filtration Slits (Slit Pores)** (~$25\ \text{nm}$ wide).
2. **Renal Tubule**: - **Proximal Convoluted Tubule (PCT)**: Located in cortex; brush-border microvilli. - **Loop of Henle**: Hairpin loop dipping into the medulla (long in juxtamedullary nephrons). - **Distal Convoluted Tubule (DCT)**: Located in cortex. - **Collecting Duct**: Drains multiple DCTs; descends through medullary pyramids to the duct of Bellini.

- **The Three Stages of Urine Formation**: 1. **Glomerular Ultrafiltration**:
  - Driven by **Net Filtration Pressure (NFP)**:

$$\text{NFP} = \text{Glomerular Hydrostatic Pressure (GHP} \sim 60) - [\text{Colloid Osmotic Pressure (BCOP} \sim 32) + \text{Capsular Hydrostatic Pressure (CHP} \sim 18)]$$

  - Produces **180 litres of protein-free glomerular filtrate per day** (GFR = $125\ \text{mL/min}$). 2. **Selective Tubular Reabsorption**:
  - Over **99% of the 180-litre filtrate is reabsorbed** back into peritubular capillaries, leaving only **~ 1.5 litres of urine** per day.
  - Glucose, amino acids, and Na^+ are reabsorbed via high-affinity active transport; water follows passively via osmosis. 3. **Tubular Secretion**:
  - Active transport of waste ions (H^+, K^+, NH_4^+), creatinine, and drugs (penicillin, sulphonamides) from peritubular capillaries directly into the tubular lumen; vital for maintaining blood pH (**7.35–7.45**) and electrolyte balance.

- **Counter-Current Multiplier & Concentrator**:
- The parallel counter-directional flow of tubular fluid in the descending and ascending limbs of Henle's loop, combined with counter-current blood flow in the **Vasa Recta** capillaries, establishes a steep hyperosmotic interstitial gradient in the renal medulla (from $300\ \mathrm{mOsm/L}$ in cortex to $\mathbf{1200\ mOsm/L}$ in deep renal papilla).
- Enables humans to concentrate urine up to **4 times higher** than normal blood plasma osmolarity (~$1200\ \mathrm{mOsm/L}$), conserving critical body water.
- **Hormonal Feedback Loops Governing Excretion**:
- **Renin-Angiotensin-Aldosterone System (RAAS)**:
  - Trigger: Drop in GFR, renal blood flow, or arterial pressure.
  - **Juxtaglomerular Cells (JG Cells)** of afferent arterioles secrete the proteolytic enzyme **Renin**.
  - Renin cleaves plasma **Angiotensinogen** (made by liver) $\rightarrow$ **Angiotensin-I**.
  - **Angiotensin Converting Enzyme (ACE)** in lung capillaries converts Angiotensin-I $\rightarrow$ **Angiotensin-II**.
  - Angiotensin-II is a potent arterial vasoconstrictor that elevates blood pressure and stimulates the adrenal cortex to secrete **Aldosterone**.
  - Aldosterone stimulates active Na^+ and water reabsorption in the DCT and collecting duct, restoring blood volume.
- **Antidiuretic Hormone (ADH / Vasopressin)**:
  - Trigger: Hypothalamic osmoreceptors detect dehydration (blood osmolarity $> 300\ \mathrm{mOsm/L}$).
  - Synthesised in hypothalamus, released from posterior pituitary; acts on collecting ducts to promote rapid water reabsorption.
- **Atrial Natriuretic Factor (ANF)**:
  - Secreted by myocardial cells of cardiac atria when blood pressure is excessively elevated; acts as a powerful vasodilator and stimulates excretion of sodium (Na^+) and water, directly opposing RAAS.

One-Liner Revision -- Digestion, Respiration and Excretion

#	Fast Fact
1	Digestion starts in the **mouth** (salivary amylase/ptyalin digests starch)
2	Stomach secretes = HCl + Pepsin (protein digestion) + Mucin (protects lining)
3	Liver = largest **gland** of the body; also largest internal organ
4	Bile secreted by = **liver**; stored in = **gallbladder**
5	Main site of digestion and absorption = **Small intestine** (duodenum/jejunum/ileum)
6	Large intestine main function = **water absorption**; no digestion
7	Longest part of alimentary canal = **Small intestine** (~6-7 metres)
8	Pancreas = both **exocrine** (digestive enzymes) and **endocrine** (insulin/glucagon)
9	Aerobic respiration: Glucose + O2 = CO2 + H2O + **38 ATP**
10	Anaerobic in muscles = **lactic acid** (causes muscle cramp/fatigue)
11	Anaerobic in yeast = **ethanol + CO2** (fermentation; used in brewing)
12	Breathing rate (adults) = 15-18 breaths/min at rest
13	Tidal volume = air breathed in/out in one normal breath (~500 mL)
14	Vital capacity = maximum air exhaled after maximum inhalation (~4.5 L)
15	Gas exchange in lungs = in **alveoli** (O2 diffuses in; CO2 diffuses out)
16	Haemoglobin carries **O2** as oxyhaemoglobin; also carries some CO2
17	CO poisoning = CO binds Hb **240x more tightly** than O2 (cherry-red colour)
18	Mammals excrete **Urea** (ureotelic); Birds/reptiles excrete **Uric acid**
19	Fish excrete **Ammonia** (ammonotelic; highly toxic; needs lots of water)
20	Nephron = functional unit of kidney; Bowman's capsule + tubules + loop of Henle
21	Urine formation: Filtration + Selective reabsorption + Tubular secretion
22	Normal urine output = ~1-1.5 L/day; urine pH ~6 (slightly acidic)
23	Kidney stone = **calcium oxalate** (most common); treated by lithotripsy
24	Dialysis = used when kidneys fail; hemodialysis filters blood through machine
25	Skin is also an excretory organ = excretes sweat (water, NaCl, urea)
26	Lungs excrete = CO2 and water vapour
27	Liver excretes = bile pigments (bilirubin from dead RBCs)

#	Fast Fact
28	ADH (Anti-diuretic hormone) = controls water reabsorption in kidney tubules
29	Jaundice = yellow skin/eyes due to accumulation of **bilirubin**
30	Appendix = vestigial organ in large intestine; removed in appendicitis